

JOPLIN 24 N, MO, USA

WMO: 725590

Lat: 37.4277N

Lon: 94.5829W

Elev: 290

StdP: 97.89

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 23908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-11.0	-18.6	0.8	-13.0	-16.0	1.0	-10.3	8.1	11.0	7.3	12.1	2.2	N/A	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	35.5	23.6	33.6	24.0	32.2	24.1	26.0	31.9	25.4	31.3	24.8	30.5	2.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	20.0	29.6	23.6	19.1	29.0	23.0	18.4	28.3	82.1	31.9	79.2	31.3	76.6	30.5	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.2	5.4	-18.0	36.9	4.0	2.8	-20.9	38.9	-23.2	40.5	-25.4	42.1	-28.3	44.2		
			-18.8	27.1	4.1	0.7	-21.7	27.5	-24.2	27.9	-26.5	28.3	-29.5	28.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	0.7	2.7	8.8	14.0	18.7	23.9	25.6	24.7	20.9	14.3	8.0	2.7	(6)	
(7)		DBStd	10.02	6.46	6.61	6.14	4.96	4.37	2.81	2.83	3.16	3.79	5.12	5.49	5.49	(7)	
(8)		HDD10.0	979	295	219	95	19	2	0	0	0	18	97	234		(8)	
(9)		HDD18.3	2443	548	439	298	146	51	2	1	1	17	146	310	486		(9)
(10)		CDD10.0	2369	6	13	59	138	272	418	483	456	327	151	38	7		(10)
(11)		CDD18.3	791	0	0	3	14	63	170	226	198	94	21	2	0		(11)
(12)		CDH23.3	7713	0	4	29	115	512	1618	2347	2023	872	185	8	0		(12)
(13)	CDH26.7	3168	0	0	3	12	118	612	1098	960	327	37	1	0		(13)	
(14)	Wind	WSAvg	2.1	2.4	2.7	3.0	3.0	2.2	2.1	1.7	1.3	1.3	1.5	2.0	2.1	(14)	
(15)	Precipitation	PrecAvg	1211	38	49	87	126	173	163	112	78	128	90	83	64	(15)	
(16)		PrecMax	1609	82	139	216	327	353	418	405	180	455	243	247	206	(16)	
(17)		PrecMin	773	7	1	36	28	41	39	9	0	34	19	1	9	(17)	
(18)		PrecStd	225	25	36	49	65	79	78	74	42	88	57	63	47	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	23.4	26.6	28.2	30.8	34.6	38.0	38.6	34.1	29.9	24.5	20.3	(19)	
(20)		MCWB	14.1	15.4	17.7	18.3	21.9	24.5	23.0	24.2	22.9	21.8	16.2	15.7		(20)	
(21)		2%	DB	16.7	19.4	23.5	26.4	29.4	32.6	35.6	35.6	32.0	27.7	21.6	17.2	(21)	
(22)		MCWB	10.6	13.6	16.1	17.6	21.5	24.5	23.4	24.0	23.7	18.9	15.0	13.0		(22)	
(23)		5%	DB	14.1	16.7	20.9	24.6	28.0	31.5	33.7	33.7	30.5	25.4	19.5	14.2	(23)	
(24)		MCWB	9.2	12.3	13.9	17.0	20.9	23.9	24.4	23.3	22.9	17.9	14.3	9.9		(24)	
(25)		10%	DB	10.7	13.7	18.7	22.6	26.5	30.3	32.1	31.9	28.5	23.2	17.2	11.8	(25)	
(26)		MCWB	6.9	9.0	12.7	16.2	20.2	23.5	24.5	23.6	21.7	17.0	12.3	8.5		(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.9	19.0	20.8	23.9	26.3	26.9	26.7	25.2	22.7	18.6	16.4	(27)	
(28)		MDCB	19.5	21.4	25.3	26.2	29.0	32.4	32.5	32.4	31.6	28.4	21.1	19.6		(28)	
(29)		2%	WB	11.6	15.1	17.2	19.4	22.9	25.5	26.1	25.8	24.3	21.4	16.4	13.6	(29)	
(30)		MDCB	15.0	18.4	22.5	24.1	27.9	31.3	31.8	32.0	30.8	25.4	19.6	15.9		(30)	
(31)		5%	WB	9.4	12.8	15.6	18.2	21.9	24.8	25.5	25.0	23.5	20.0	14.8	11.0	(31)	
(32)		MDCB	13.1	16.1	19.7	22.9	26.6	30.4	31.2	31.2	29.5	23.1	18.4	13.2		(32)	
(33)		10%	WB	6.8	8.9	13.7	17.1	20.9	24.0	24.9	24.1	22.6	18.2	13.0	8.7	(33)	
(34)		MDCB	10.6	13.6	17.3	21.4	25.4	29.4	30.5	29.9	27.7	22.1	16.5	11.4		(34)	
(35)	Mean Daily Temperature Range		MDBR	11.4	11.3	12.0	11.8	10.8	10.7	11.3	12.0	12.3	13.0	12.5	10.6	(35)	
(36)		5% DB	MCDBR	15.7	15.9	14.7	13.7	12.6	11.7	13.6	14.7	13.5	14.7	14.8	14.5	(36)	
(37)		MCWBR	10.5	10.7	8.3	7.5	6.2	5.4	4.7	5.2	6.0	7.5	9.6	9.6	(37)		
(38)		5% WB	MCDBR	14.0	14.8	12.1	11.3	11.0	11.1	11.0	12.1	12.2	11.1	12.9	11.7	(38)	
(39)		MCWBR	10.4	11.2	7.6	7.1	6.0	5.5	5.0	5.4	5.7	6.7	9.6	9.3	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.313	0.345	0.380	0.422	0.424	0.439	0.441	0.387	0.345	0.310	0.290	(40)		
(41)		taud	2.449	2.432	2.389	2.313	2.253	2.281	2.279	2.251	2.370	2.464	2.504	2.507	(41)		
(42)		Ebn at Noon	886	921	918	900	863	859	842	832	863	873	872	880	(42)		
(43)		Edh at Noon	83	96	111	127	137	133	132	133	112	92	79	74	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.39	2.99	3.97	5.04	5.67	6.51	6.48	5.81	4.91	3.57	2.54	2.03	(44)		
(45)		RadStd	0.21	0.46	0.26	0.35	0.47	0.52	0.40	0.37	0.36	0.42	0.28	0.23	(45)		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page